Can intrapartum Sildenafil Citrate safely avert the risks of contraction-induced hypoxia in labour? iSEARCH – a pragmatic multicentre Phase III randomised controlled trial.

STATISTICAL ANALYSIS PLAN

Draft Version 1.2

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Study identifiers:

NHMRC CTC protocol number: CTC0303

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Chief Investigator & Study Chair: Professor Sailesh Kumar

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Coordinating Centre: NHMRC Clinical Trials Centre, The University of Sydney.

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1 Administrative Information

1.1 Study Identifiers

- NHMRC CTC protocol number: CTC0303
- ANZCTR: ACTRN12621000231842

1.2 Revision History

Version	Date	Changes made to documents	Authors
1.0	23rd September 2024	First version drafted	R O'Connell
1.1	14th October 2024	Second version drafted	R O'Connell
1.2	25th October 2024	Minor changes. SAP finalised.	R O'Connell

1.3 Contributors to the Statistical Analysis Plan

1.3.1 Roles and Responsibilities

Name and ORCID	Affiliation	Role on study	SAP contribution
Rachel O'Connell	NHMRC CTC	Statistician	Wrote SAP
John Simes	NHMRC CTC	Senior Trialist and Co-Investigator	Provided input for statistical methods
Sailesh Kumar	University of Queensland	Study Chair	Reviewed and approved SAP

1.3.2 Approvals

The undersigned have reviewed this plan and approve it as final. They find it to be consistent with the requirements of the protocol as it applies to their respective areas. They also find it to be compliant with ICH-E9 principles and, in particular, confirm that this analysis plan was developed in a completely blinded manner (i.e. without knowledge of the effect of the intervention(s) being assessed).

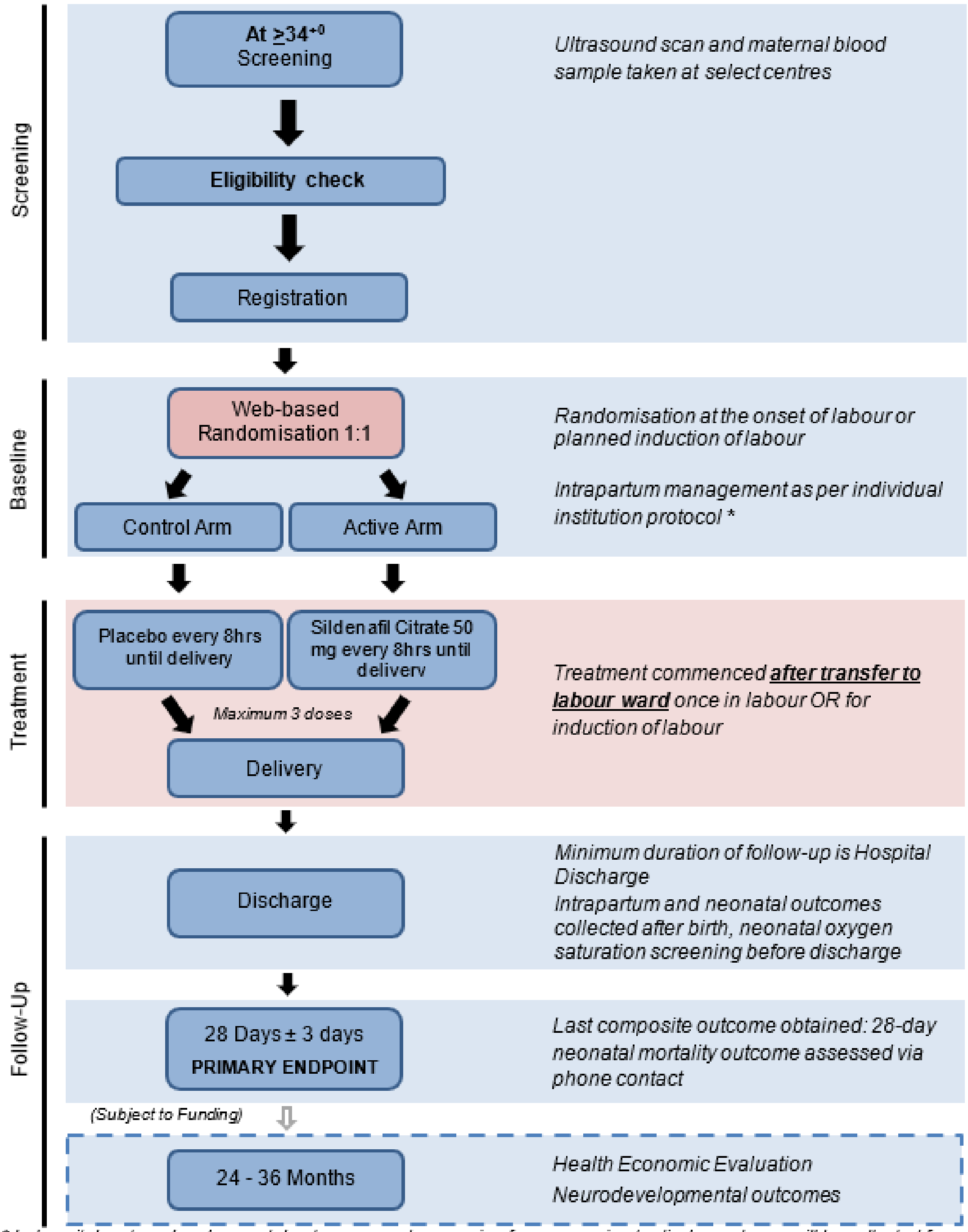
Sailesh Kumar 25/10/2024

Rachel O'Connell 25/10/2024

2 Study Synopsis

2.1 Study Schema

Duration of accrual: 4 years; **Stratification:** Study Sites; **Primary Endpoint:** Composite of ten perinatal outcomes related to intrapartum hypoxia, determined 28 days from birth.



** In-hospital maternal and neonatal outcome events occurring from screening to discharge home will be collected from medical records. There are no formal study assessments visits.*

Figure 1: Trial Schema

2.2 Aim and Objectives

The aim of this study is to test the hypotheses that ≤3 doses of oral 50 mg SC vs. placebo will improve perinatal and/or maternal outcomes related to intrapartum hypoxia.

The primary objective is to demonstrate or refute a reduction of 35% in the relative risk of the composite perinatal outcome (from 7% to 4.55%) comprising ten components, shown below and in Table 1. Composite neonatal outcomes, appropriately constructed, are often employed in multicentre trials.

Composite primary outcome:

I. Intrapartum stillbirth, or
II. 28-day neonatal death, or
III. Apgar score <4 at 5 minutes, or
IV. Cord artery pH <7.0, or
V. Neonatal encephalopathy, or
VI. Neonatal seizures, or
VII. Neonatal respiratory support lasting over 4 h, or
VIII. Neonatal unit (Special Care or Intensive care) admission lasting over 48 h, or
IX. Persistent pulmonary hypertension of the newborn, or
X. Meconium aspiration

Table 1

Components of the primary outcome	Rate*	Association with long term adverse outcome
Intrapartum stillbirth [45]	0.1%	-
28 day neonatal mortality[46]	0.24%	-
Apgar score <4 at 5 minutes	0.5%	↑ risk of cerebral palsy[47]
Umbilical Cord artery pH <7.0	2.2%	↑ risk of HIE[48]
Neonatal encephalopathy	0.5%	↑ risk of death or disability[49]
Neonatal seizures	0.25%	↑ risk of death or disability[498]
Neonatal respiratory support for >4 h	3.6%	↑ risk of cerebral palsy[50]
Neonatal unit admission for >48 h	4.3%	↑ asthma after term respiratory morbidity[51]
Persistent pulmonary hypertension	0.04%	↑ risk of death or disability[52]
Meconium aspiration	0.75%	↑ risk of death or disability[53]
Total (corrected for overlap)	7.0%	

**based on data from Grobman et al,[54] and the Australia New Zealand Neonatal Network[55]*

Secondary objectives:

1. Reduces the relative risk of each individual component of the composite primary outcome (detailed in Table 1).
2. Reduces the relative risk of CS or IVB for fetal distress by 25% (from 20% to 15%).
3. Is cost-effective.

Tertiary objectives:

1. CS for suspected fetal distress
2. IVB for suspected fetal distress
3. CS for other indications
4. IVB for other indications
5. Spontaneous vaginal birth
6. Length of 1st and 2nd stage of labour
7. Post-partum haemorrhage (blood loss >1000ml)
8. Post-partum hysterectomy
9. Uterine rupture
10. Maternal Intensive Care Unit admission

11. Maternal mortality
12. Reduces the relative risks of cerebral palsy, cognitive delay and neurosensory disability at 2 years of age.

The correlative objectives of this study in a subset of participants are:

1. To determine if any fetal biophysical parameters (estimated fetal weight, umbilical artery pulsatility index, middle cerebral artery pulsatility index or cerebroplacental ratio) are predictive of efficacy of SC treatment.
2. To determine if maternal PlGF, sFlt-1 or sFlt-1:PIGF ratio are predictive of efficacy of SC treatment.

The analysis of the 2-year outcomes, correlative objectives and the economic evaluation is outside the scope of this analysis plan.

2.3 Design

This study is a randomised, multicentre, two-arm parallel, phase III, double-blind, placebo-controlled trial.

2.4 Study population

Participants must meet all of the inclusion criteria and none of the exclusion criteria to be eligible for this trial. No exceptions will be made to these eligibility requirements at the time of registration. All enquiries about eligibility should be addressed by contacting the NHMRC CTC prior to registration.

2.4.1 Target Population

Pregnant women from 34^{+0} weeks planning a vaginal birth at term (≥37 weeks gestation). The research will be conducted in a manner sensitive to the cultural principles of Aboriginal and Torres Strait Islander communities.

2.4.2 Inclusion Criteria

1. Women with singleton or dichorionic twin pregnancies planning in-hospital vaginal birth at term (≥37 weeks gestation).
2. Age 18 years and older.
3. Willing and able to comply with all study requirements.
4. Signed, written informed consent. For Non-English speaking participants, a Health Care Interpreter must be engaged to obtain informed consent.

2.4.3 Exclusion Criteria

1. A woman should not be entered if the responsible clinician or the woman are, for any medical or non-medical reasons, reasonably certain that SC **would be inappropriate** for her in comparison with no treatment or some other treatment that could be offered in or outside the trial.
2. Triplets or higher order multiple births, which are generally delivered electively before term.
3. Contraindications to the investigational product SC (see 5.4).
4. Participants who are taking any type of nitrate drug therapy or who utilize short-acting nitrate-containing medications during labour [such as sodium nitroprusside, bosentan, fosamprenavir and ritonavir combination, hepatic enzyme inhibitors CYP3A4 (including itraconazole, ketoconazole, ritonavir, cimetidine, erythromycin, saquinavir, darunavir), or hepatic enzyme substrates (CYP3A4), medications used to treat pulmonary arterial hypertension, and other phosphodiesterase type 5 inhibitors like riociguat], due to the risk of potentially life-threatening hypotension.
5. Severe hepatic or renal impairment.

2.5 Treatment Plan

Sildenafil Citrate is the study intervention in this trial. Placebo is the control intervention. Treatment will commence after transfer to the labour ward once in labour or for induction of labour. The dose of SC is 50mg orally or equivalent placebo every 8 hours to a maximum of three doses.

2.5.1 Study Treatment

Participants will be randomised to one of the following two arms:				
Arm	Agent(s)	Dose	Route	Duration
Active	Sildenafil Citrate	50 mg	Oral	Every 8 hours from the start of labour or induction of labour to a maximum of 3 doses.
Control	Placebo	Placebo	Oral	

2.5.2 Required Background Treatment

All participants are expected to receive background treatment consistent with routine obstetric care.

2.6 Randomisation and Blinding

Women will be centrally randomised through the NHMRC CTC. Treatment allocation will be balanced using minimisation for the study site and women will be allocated to the treatment groups in a ratio of 1:1.

Registration, including randomisation, must be done according to the instructions in the Study Manual.

Randomisation will be performed after registration and consent to participate has been obtained when the participant is admitted to the hospital either in spontaneous labour or for induction of labour. It will be done according to the instructions in the Study Manual and should be undertaken after all screening assessments have been performed and the responsible investigator has verified the participant's eligibility. Women may only be randomised once in this trial. Once the randomisation process has been completed, the participant will be assigned a treatment arm.

The study is double-blinded. Unblinding is not generally necessary for the management of a participant with an adverse event and, as it has an impact on the study's validity, is strongly discouraged. The need for unblinding should be very uncommon as the study intervention is rarely associated with severe side effects and it will not delay or prevent standard management of the participant. However, if required, it can be done after discussion with the Study Chair and NHMRC CTC Clinical Lead using the emergency unblinding number in the study manual.

2.7 Outcome Definitions

In-hospital perinatal and maternal outcome events occurring from screening to discharge home will be collected from medical records. With the exception of 28-day neonatal mortality, all outcome data will be routinely available before discharge. There are no formal study assessments visits. Research midwives will contact participants 28 days after discharge to ascertain if there any issues relating to the study that require additional follow up.

2.7.1 Primary Composite Outcome

A composite of the following adverse perinatal outcomes:

- Intrapartum stillbirth or
- 28-day neonatal mortality or
- Apgar score <4 at 5 minutes or
- Cord artery pH <7.0 or
- Neonatal encephalopathy (Sarnat score) up to 28 days or
- Neonatal seizures up to 28 days or

- Neonatal respiratory support >4h or
- Neonatal unit (Special Care or Intensive Care) admission lasting >48h or
- Persistent pulmonary hypertension of the newborn or
- Meconium aspiration

2.7.1.1 Intrapartum Stillbirth or 28-day Neonatal Death

Intrapartum stillbirth is the death in labour of an infant that was alive at the commencement of labour.

2.7.1.2 28-day neonatal mortality

Neonatal mortality is the death of an infant within 28 days of birth

2.7.1.3 Apgar score <4 at 5 minutes

Numerical score to evaluate the infant's condition at 5 minutes after birth based on five characteristics – heart rate, respiratory condition, muscle tone, reflexes and colour. Assessed routinely and recorded in the medical record.

2.7.1.4 Cord artery pH <7.0

Significant metabolic acidosis defined as cord arterial blood pH <7.0

2.7.1.5 Neonatal encephalopathy

A clinically defined syndrome of disturbed neurologic function in the earliest post-natal days of life in an infant born at or beyond 35 weeks of gestation, manifested by a subnormal level of consciousness or seizures, and often accompanied by difficulty with initiating and maintaining respiration and depression of tone and reflexes. Measured in infants >35 weeks using a modified Sarnat criteria to assess severity of encephalopathy and graded from 0-3. Grade 0, no hypoxic ischaemic encephalopathy to grade 3, severe hypoxic ischaemic encephalopathy.

2.7.1.6 Neonatal seizures

Defined as the paroxysmal alteration in neurological function of behavioural, motor or neurological function. May be clinical or diagnosed on electroencephalogram.

2.7.1.7 Respiratory support lasting >4 hours

Supplementary respiratory support includes intubation and intermittent positive pressure ventilation, high flow oxygen via nasal cannula or CPAP.

2.7.1.8 Neonatal unit admission lasting >48 hours

Neonatal unit admission is defined as the number of hours from admission to discharge.

2.7.1.9 Persistent pulmonary hypertension of the newborn

Persistent pulmonary hypertension of the newborn is defined as a syndrome of failed circulatory adaptation at birth due to delay or impairment in the normal fall in pulmonary vascular resistance that occurs following birth. Infants with oxygen saturations of >95% are very unlikely to have persistent pulmonary hypertension of the newborn.

2.7.1.10 Meconium Aspiration

Presence of meconium in the trachea and symptoms/signs fulfilling criteria for persistent pulmonary hypertension of the newborn

2.7.2 Secondary Outcomes

2.7.2.1 Individual components of primary composite endpoints

These include individual components of the primary composite endpoint (Intrapartum stillbirth, 28-day neonatal mortality, Apgar score <4 at 5 minutes, Cord artery pH <7.0, Neonatal encephalopathy, Neonatal seizures, Neonatal respiratory support >4h, Neonatal unit (Special Care or Intensive Care) admission lasting >48h, Persistent pulmonary hypertension of the newborn or Meconium aspiration) of the primary composite outcome.

2.7.2.2 CS or IVB for fetal distress

Defined as reduction in the relative risk of CS or IVB for fetal distress by 25% (from 20% to 15%).

2.7.3 Tertiary Outcomes

2.7.3.1 CS for fetal distress

Defined as an emergency CS for fetal distress during labour as detected by fetal heart rate abnormalities based on the Royal Australian and New Zealand College of Obstetricians and Gynaecologists guidelines and/or abnormal fetal scalp pH or lactate levels.

2.7.3.2 Instrumental vaginal birth for fetal distress

Defined as a vaginal birth using forceps or vacuum extraction for fetal distress.

2.7.3.3 CS for other indications

Other indications include CS for failure to progress, failed induction of labour, malpresentation, failed instrumental birth, cord prolapse/presentation, uterine rupture etc.

2.7.3.4 Instrumental vaginal birth for other indications

Defined as a vaginal birth using forceps and vacuum extraction for non-fetal distress indications.

2.7.3.5 Spontaneous vaginal birth

Defined as the uncomplicated vaginal birth of an infant.

2.7.3.6 Length of 1st and 2nd stage of labour

The 1^{st} stage of labour is defined as commencing when labour is diagnosed by the attending obstetric team to full dilatation (i.e. 10cm). The 2^{nd} stage of labour is defined as the period of time from full cervical dilatation to birth of the infant.

2.7.3.7 Post-partum haemorrhage

Defined as excessive bleeding (>1000ml) following the birth of a baby.

2.7.3.8 Post-partum hysterectomy

Defined as the removal of the uterus (+/- cervix) after birth.

2.7.3.9 Uterine rupture

Defined as a full thickness tear through the uterine myometrium and serosa.

2.7.3.10 Maternal Intensive Care Unit admission

Intensive care unit admission is defined as the admission of the participant to an intensive care unit.

2.7.3.11 Maternal Mortality

Maternal mortality is defined as death of mother before discharge

2.8 Sample Size

The incidence of the composite adverse outcome is estimated at 7% based on data in a recent large RCT in term pregnancies and the Australian and New Zealand Neonatal Network. To detect a 35% reduction from 7% to 4.55% for an alpha of 0.05 and >80% power with a 10% drop-out in each arm, needs 3,200 women, also yielding >90% power to detect a 25% reduction for the secondary outcome (CS or IVB for fetal distress).

2.9 Interim Analysis

The Independent Data and Safety Monitoring Committee (IDSMC) will review interim data on the primary outcome, specified adverse events and other evidence after 50% of recruitment or whenever they deem appropriate informed by recommendations of Peto, Pocock and others. There will be no adjustment to alpha for interim analyses using a Haybittle-Peto boundary.

Interim analyses of the primary composite outcome: The IDSMC will advise the TMC if in their view there is proof beyond reasonable doubt of net benefit or harm for the primary composite outcome, employing a commonly used formal threshold of P<0.001 for nominal significance, as recommended by Geller and Pocock.

Interim analyses of mortality: The IDSMC will advise the TMC if in their view there is a difference in mortality due to intrapartum stillbirth and/or 28-day neonatal mortality identified as a deviation from the null indicated by a Haybittle-Peto boundary which is equivalent to P <0.0027, which would be needed to justify recommending early stopping.

3 Statistical Analysis

3.1 Analysis datasets

3.1.1 Modified Intention To Treat population

The principal analysis of study endpoints will be according to the modified intention-to-treat (mITT) principle including all subjects with available outcome data. All randomised babies who received no study treatment and were later found to be ineligible (before randomisation) will be referred to independent adjudicators for a blinded assessment. If the reason is judged not related to the randomised treatment allocation or any outcome events post randomisation the baby will be excluded from the mITT dataset in a sensitivity analysis.

3.1.2 Safety population

All randomised patients who receive at least 1 dose of study treatment will constitute the safety population. Patients will be counted in the group according to the treatment they received.

3.2 General Principles

Summary statistics (means and standard deviations, medians and inter-quartile range) will be presented for continuous baseline variables. Number (%) will be presented for categorical variables. Categorical variables will be compared with chi-squared tests, with a test for trend over ordered categories (Mantel-Haenszel Chi-square). Continuous variables will be compared using t-tests or the Wilcoxon rank-sum test if non-normally distributed.

Analyses of the primary and secondary outcomes will adhere to the intention-to-treat principal using Generalized Estimating Equations (GEE) to accommodate correlation of data between siblings from multiple births (log-binomial). An exchangeable correlation structure will be assumed. As the data is sparsely clustered due to a small proportion of mother's having twins (i.e., ~0.6%), methods which treat all babies as independent observations will also be performed. Results from the two methods will be compared. Failure to take the non-independence of twins into account can lead to under-estimated standard errors and biased parameter estimates. Therefore, the GEE method is intended to be the primary analysis provided that the models are numerically stable and converge well. Maternal secondary/tertiary endpoints will be analysed using Generalised Linear Models (log-binomial).

Intervention effect will be presented as relative risk (RR) or mean difference (MD), as appropriate, with 95% confidence intervals. Two-tailed P values <0.05 will be considered statistically significant. Primary analyses will be unadjusted. Numbers needed to treat to prevent one adverse outcome will be calculated. Secondary analyses of the primary outcome will be carried out including adjustment for relevant baseline prognostic factors using multiple (log-binominal or Robust Poisson) regression. If appropriate, the conditional binomial exact test will be used to compare groups when expected cell counts are small (i.e., <5).

The key secondary outcomes for iSEARCH comprise the individual components of the primary composite outcome (Section 2.7.1 above). P values adjusted for the ten comparisons performed for this set of endpoints will be derived using the Benjamin-Hochberg procedure, with a family-wise error rate of 5%. Results of other endpoints, subgroup and sensitivity analyses will be interpreted in proper context and with due consideration of the risk of Type I error. In general, complete-case analysis will be performed with no imputation done for missing data.

3.3 Subject Disposition

A patient disposition flow-chart will be presented showing the number of mothers screened, eligible, consented and randomised to each treatment arm. Reasons for withdrawal, losses to follow-up and numbers available for mITT and safety analyses will be detailed.
A table will also be prepared showing the number of randomised mothers found to be ineligible with reasons detailed.

3.4 Treatment Adherence

The proportions of mothers that received treatment will be calculated. The numbers receiving 1, 2 and all 3 doses will be described along with the time from randomisation to each dose and the dose received. Reasons for not giving treatment will also be detailed.

3.5 Baseline Analyses

A table will be prepared showing the stratification factors by treatment arm (i.e., site). Descriptive statistics will be prepared to summarise baseline characteristics (i.e., maternal demographics and clinical details) by treatment allocation.

3.6 Analysis of the Primary Outcome

The primary analysis of the primary outcome a composite of ten perinatal outcomes will be based on a GEE log-binomial model. A relative risk, with a 95% confidence interval (CI) will be presented to indicate the relative reduction (or increase) in risk associated with Sildenafil therapy along with numbers and proportions meeting the outcome in each treatment arm. Numbers needed to treat to prevent one adverse outcome will be calculated.

In 27% of babies a cord artery test for pH level was not performed. As this is a component of the primary outcome 'not done' for this component will be assumed to be no event.

3.7 Analysis of Secondary and Tertiary Outcomes

The analysis of neonatal secondary outcomes will be evaluated with the use of GEE log-binomial models. Maternal outcomes will be analysed using a log-binomial generalized linear model for binary outcomes, and a t-test for continuous outcomes.

3.8 Safety Outcomes

An analysis of neonatal and maternal mortality will also be performed by treatment received. The frequency of maternal adverse events of interest will be analysed by treatment received. The comparison between treatment arms will employ a chi-squared test or the conditional binomial exact test if expected cell counts are small (i.e., <5).

3.9 Adjusted Analyses

Secondary analysis of the primary endpoint to adjustment for prognostic variables will be carried out using multivariable regression. Adjustment variables will include: (1) Small for gestational age (Birth weight <10th centile for gestation); (2) Pre-eclampsia; (3) Method of delivery for previous pregnancy (no prior delivery, spontaneous vaginal delivery, instrument assisted vaginal delivery, and caesarean section); and (4) post-natal diagnosis of fetal anomaly. If the log-binomial model fails to converge a Robust (modified) Poisson regression model or other computational method will be applied and/or the clustering amongst twins will not be accounted for. If it is not possible to fit a GEE model, multiple birth will be included as a covariate in the model.

3.10 Sensitivity Analysis and missing data

The following sensitivity analyses will be performed for the primary outcome:

1) Imputation of missing data will be performed for Cord artery pH level using maximum likelihood estimation based on the observed pooled event rates within twins (c_T) and non-twins (c_{NT}). The imputation will be stratified by twin status due to the much higher Cord Artery pH <7 event rate among twins (11% vs 0.6% among babies where cord artery pH was tested) and higher missing data rate. This calculation will take into account the occurrence of the other 9 components of the composite primary outcome. The imputation will be performed according to the following steps:

   a. The Cord Artery pH <7 event rates (c_T and c_{NT}) will be calculated among babies where Cord Artery pH testing was performed,
   b. Of those babies having an event the fractions (f_T and f_{NT}) that suffered none of the other 9 components will be calculated,
   c. The additional number of events will be calculated by multiplying the number of babies with Cord Artery Not Done who had not already met the primary endpoint by p and f above within each treatment arm (S_T, S_{NT}, P_T, P_{NT}) e.g., for non-twins in Placebo arm the calculation is $P_{NT}c_{NT}f_{NT}$.
2) An analysis of the primary outcome based on 9 components only will be performed with Cord artery pH <7 excluded from the definition.
3) A generalised linear model (log-binomial) will be fitted ignoring the correlation between twins.
4) Models will be fitted with an extra covariate included to indicate whether or not Cord Artery testing was done in order to adjust for any imputation. A GLM model will be used if there are numerical estimation problems with the GEE model.

3.11 Subgroup Analyses

Consistency of the treatment effect on the primary end point will be evaluated across the following pre-specified subgroups: (1) Small for gestational age (Birth weight <10th centile for gestation); (2) multiple pregnancy; (3) pre-eclampsia; (4) Method of delivery for previous pregnancy CS; (5) non-cephalic presentation; (6) post-natal diagnosis of fetal anomaly.

This will be performed by fitting an interaction term between the subgroup of interest and treatment in a log-binomial GLM model (along with the relevant main effect terms). The two-sided level of significance is 0.05, however due to the number of analyses results should be interpreted with caution. Treatment effects will be determined from a model fitted to each of the levels of the subgroup, e.g., for pre-eclampsia separate models will be fitted for yes and no. Models will not include adjustment for baseline covariates. A GEE model will not be applied because the number of twins in each subgroup maybe too small for models to converge.

4 Appendix I: Proposed Tables and Figures

4.1 Summary of Patient Population

Figure 2: Patient Flow Chart

Table 1: Summary of patient eligibility by randomized treatment arm.

Characteristic	**Sildenafil Citrate (N =)**	**Placebo (N =)**
Total randomized, N		
Eligible, n (%)		
Ineligible, n (%)		
Reason 1, n		
Reason 2, n		
….		

Table 2: Stratification factors by randomised treatment arm. Data are n (%).

Characteristic	**Sildenafil Citrate (N =)**	**Placebo (N =)**
Site:		
Casey Hospital		
Dandenong Hospital		
Gold Coast University Hospital		
Ipswich Hospital		
John Hunter Hospital		
Mater Mothers' Hospital		
Monash Medical Centre - Clayton		
Royal Brisbane and Women's Hospital		
Royal Hospital for Women		
Royal Prince Alfred Hospital		
Sunshine Coast University Hospital		
Sunshine Hospital		
The Royal Women's Hospital		

Table 3: Maternal demographics and clinical details by randomised treatment arm. Data are n (%) unless otherwise indicated.

	Sildenafil Citrate (N =)	**Placebo (N =)**
Maternal details:		
Mother's Age (yrs, mean [SD])		
Gestation at randomisation (weeks+day, mean [SD])		
Ethnicity:		
Aboriginal/Torres Strait Islander		
Australia/New Zealand		
African		
Middle Eastern		
European		
South-east Asian		
North-east Asian		
South Asian		
Pacific		
North or South American		
Other or mixed ethnicity		
Booking BMI (kg/m^2, mean [SD])		
Has the mother smoked during this pregnancy:		
No		
Yes, ≥ 10 cigarettes a day		
Yes, < 10 cigarettes a day		
Obstetric History:		
Parity (births > 20 weeks gestation):		
0		
1		
2		
3		
4+		
Method of delivery last pregnancy:		
Spontaneous vaginal delivery		
Instrumental assisted vaginal delivery		
Elective caesarean section		
Emergency caesarean section		
Indication of previous emergency Caesarean Section:		
Fetal distress		
Failure to progress		
Other		
Mode of conception for this pregnancy:		
Natural conception		
Assisted conception (IVF, Ovulation Induction, etc.)		
Medical history:		
Current high blood pressure:		
No		
Yes		

	Sildenafil Citrate (N =)	Placebo (N =)
Please specify:		
Pre-eclampsia		
Pregnancy induced hypertension		
Chronic/essential hypertension		
Current diabetes:		
No		
Yes		
Type of diabetes:		
Type 1		
Type 2		
GDM		
Treatment for diabetes:		
Diet control		
Metformin		
Insulin		
Metformin and Insulin		
Other complications for this pregnancy:		
No		
Yes		
Threatened preterm labour:		
No		
Yes		
Antepartum haemorrhage:		
No		
Yes		
Fetal Growth Restriction/Small for Gestational Age:		
No		
Yes		
Other:		
No		
Yes		

506 *Table 4: Maternal current medications by randomised treatment arm. Data are n (%).*

	Sildenafil Citrate (N =)	Placebo (N =)
Any blood pressure medications:		
No		
Yes		
Alpha-blockers (or for bladder control):		
No		
Yes		
Nitrates, amyl nitrate, nicorandil (Ikoren), glyceryl trinitrate (Anginine, Lycinate, Minitran, Nitro-Dur, Nitrolingual, Transiderm) or isosorbide salts (Fenofibrate, Ifosfamide, Isosorbide Dinitrate, Isosorbide Mononitrate), Sodium nitroprusside or Bosentan (Tracleer)		
No		
Yes		
Cimetidine (Magicul, Tagamet):		
No		
Yes		
Any medication for pulmonary hypertension or PDE-5 Inhibitors:		
No		
Yes		
Antibiotics:		
No		
Yes		
Oral antifungal medicines:		
No		
Yes		
HIV protease inhibitors:		
No		
Yes		
Was aspirin taken during the pregnancy?		
No		
Yes		
When was aspirin commenced?		
At or before 16 weeks		
After 16 weeks		
What was the aspirin dosage?		
100mg		
150mg		
Other		
Other medications:		
No		
Yes		

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4.2 Compliance

Table 5: Isearch Medication by randomised treatment arm. Data are n (%) unless otherwise indicated.

	Sildenafil Citrate (N =)	Placebo (N =)	P-value*
Dose 1			
Was dose 1 given?			
No			
Yes			
Hours since randomisation given (median [range])			
Dose given:			
50 mg			
…			
Specify reason not given:			
Clinician discretion			
Participant refusal			
Infant delivered			
Other			
Dose 2			
Was dose 2 given?			
No			
Yes			
Hours since randomisation given (median [range])			
Dose given:			
50 mg			
…			
Specify reason not given:			
Clinician discretion			
Participant refusal			
Infant delivered			
Other			
Dose 3			
Was dose 3 given?			
No			
Yes			
Hours since randomisation given (median [range])			
Dose given:			
50 mg			
…			
Specify reason not given:			
Clinician discretion			
Participant refusal			
Infant delivered			
Other			
Number of doses received:			
1			
2			
3			

*Chi-squared test for categorical variables (with linear trend test for ordered variables); Wilcoxon rank-sum test for continuous variables.

4.3 Labour, Delivery and Neonatal Details

Table 6: Labour and Delivery details by randomised treatment arm. Data are n (%) unless otherwise indicated.

	Sildenafil Citrate (N =)	Placebo (N =)	P-value*
Maternal Admission			
Length of hospital stay (median [range])			
Onset of labour – n (%)			
Spontaneous			
Induction of labour (IOL)			
Gestation at onset of labour (weeks+day, mean [SD])			
Is this a twin pregnancy			
Indication for Induction of Labour:			
Fetal Growth Restriction or Small for Gestational Age			
Suboptimal fetal growth			
Large for gestational age			
Diabetes Mellitus			
Hypertension			
Maternal age			
Fetal abnormality			
Prolonged Rupture of Membranes			
Other			
Method for Induction of Labour:			
Balloon cervical dilatation			
Prostaglandin			
ARM and Syntocinon			
ARM only			
Details of Labour:			
Meconium stained liquor			
Analgesia in labour – n (%)			
None			
Epidural/spinal			
IM Pethidine or Morphine			
Nitrous Oxide (Gas and Air)			
Electronic fetal monitoring in labour			
Length of stage 1 (minutes, median [Q1, Q3])			
Length of stage 2 (minutes, median [Q1, Q3])			
Complications during labour - Yes:			
Suspected Abruption			
Intrapartum Bleeding			
Cord prolapse			
Clinical Chorioamnionitis			
Maternal eclampsia/seizures			
Other (please specify)			
Post-partum hemorrhage defined as blood loss greater than 1000mls			

	Sildenafil Citrate (N =)	Placebo (N =)	P-value*
Approximation of blood volume lost:			
1000-1499ml			
1500-1999ml			
2000ml or greater			

*Chi-squared test for categorical variables; t-test or Wilcoxon rank-sum test for continuous variables.

Table 7: Neonatal Delivery details by randomised treatment arm. Data are n (%) unless otherwise indicated.

	Sildenafi Citrate (N =)	Placebo (N =)	P-value*
Mode of birth – n (%):			
Caesarean Section			
Normal Vaginal Delivery			
Assisted breech delivery			
Instrumental vaginal birth - forceps			
Instrumental vaginal birth - vacuum			
Was this an emergency Caesarean Section?			
Primary Indication for Caesarean Section – n (%):			
Fetal distress			
Failure to progress			
Failed IOL			
Failed instrumental			
Malpresentation			
Cord prolapse/presentation			
Uterine rupture			
Other (please specify)			
Was fetal distress a factor in indication for Caesarean Section?			
Category of Caesarean Section – n (%):			
Cat 1			
Cat 2			
Cat 3			
Primary indication for Instrumental Vaginal Birth (Forceps OR Vacuum) – n (%):			
Fetal distress			
Prolonged Second Stage			
Maternal Exhaustion			
Other (please specify)			
Was fetal distress a factor for Instrumental Vaginal Birth?			
Was fetal distress reported at any point in labour or delivery?			
Fetal blood sampling done			
Was the recorded pH 7 or less or was lactate 4.8nmol/L or greater?			

*Chi-squared test

Table 8: Neonatal details by randomised treatment arm. Data are n (%) unless otherwise indicated.

	Sildenafil Citrate (N =)	**Placebo (N =)**	**P-value***
Is this a live birth?			
No			
Yes			
Was this intrapartum stillbirth?			
No			
Yes			
Gestational age (weeks+day, mean [SD])			
Gender:			
Female			
Male			
Birthweight (g, mean [SD])			
Fetal anomalies?			
Please specify:			
Major heart abnormalities (excluding simple ventricular septal defects)			
Facial clefts			
Congenital diaphragmatic hernia			
Spina bifida (including anencephaly)			
Anterior abdominal wall defects (gastroschisis or exomphalos)			
Major limb abnormalities (excluding talipes)			
Other			

*Chi-squared test (or exact test if numbers small) for categorical variables; t-test for continuous variables.

4.4 Safety Outcomes (Treatment-received analysis)

Table 9: Safety Analysis: Mortality by treatment received. Data are n (%).

	Sildenafil Citrate (N =)	**Placebo (N =)**	**P-value***
Intrapartum stillbirth			
28-day neonatal death			
Any baby death			
Primary cause of neonatal death:			
Hypoxic Ischaemic Encephalopathy			
Sepsis			
Major congenital malformation			
Metabolic syndrome			
Other			
Maternal death within 28 days			
Primary cause of maternal death:			
Haemorrhage			
Sepsis			
Pre-eclampsia			
Thromboembolism			
Other			

*Conditional binomial exact test.

Table 10: Safety analysis: Maternal adverse events of interest by treatment received. Data are n (%).

	Sildenafil Citrate (N =)	**Placebo (N =)**	**P-value***
More frequent:			
Diarrhoea			
Dizziness			
Flushing			
Gastro-oesophageal reflux/dyspepsia			
Headache			
Myalgia/limb pain			
Vision changes (blurred, increased sensitivity)			
Vomiting			
Vaginal itch			
Less frequent:			
Bronchitis			
Cellulitis/skin rash			
Epistaxis			
Hypo/paraesthesia			
Migraine-like headache			
Tremor			

*Chi-squared test (or exact test if numbers are small)

4.5 Efficacy Outcomes (Intention to Treat Analysis)

Table 11: Composite primary endpoint (and components) by randomised treatment arm. Numbers are n/N (%).

	Sildenafil Citrate (N =)	Placebo (N =)	Relative Risk* (95% CI)	P-value	NNT	Adjusted P-values (Benjamin-Hochberg)
Composite primary endpoint (10 components)						-
Intrapartum stillbirth						
28-day neonatal death						
Apgar score <4 at 5 minutes						
Cord artery pH < 7.0‡						
Neonatal encephalopathy (sarnat grade 2 or 3) within 28 days						
Neonatal seizures within 28 days						
Neonatal unit (Special Care or Intensive Care) admission >48 hours						
Neonatal respiratory support >4hrs						
Persistent pulmonary hypertension of the newborn						
Meconium aspiration						

*GEE log-binomial
‡Not Done for Cord Artery pH<7 assumed to be No.

Table 12: Multivariable analyses of the composite primary perinatal endpoint*

	Parameter Estimate	Relative Risk (95% CI)	P-value	P-value (Overall)
Intercept				
Treatment:				
Placebo				
Sildenafil Citrate				
Small for gestational age†:				
Yes				
No				
Multiple pregnancy:				
Yes				
No				
Pre-eclampsia:				
Yes				
No				
Method of delivery last pregnancy:				
No prior delivery				
Spontaneous vaginal delivery				
Instrumental assisted vaginal delivery				
Caesarean section				
Post-natal diagnosis of fetal anomaly:				
Yes				
No				

*Robust Poisson model
†Birthweight<10th centile for gestation.

Table 13: Sensitivity analyses of the composite primary perinatal endpoint

	Sildenafil Citrate (N =)	Placebo (N =)	Relative Risk (95% CI)	P-value*
Composite primary endpoint with cord artery pH <7 imputed (10 components)*				
*Cord artery pH < 7.0 (after imputation)**				
Composite primary endpoint (9 components)*				
GLM (log-binomial) – no cluster adjustment				
Composite primary endpoint (10 components) – includes adjustment for cord artery Not Done*				
Composite primary endpoint with cord artery pH <7 imputed (10 components) – includes adjustment for cord artery Not Done*				

*GEE log-binomial model

Table 14: Other Secondary and Tertiary Outcomes.

	Sildenafil Citrate (N =)	Placebo (N =)	Relative Risk /Difference in Means (95% CI)	P-value*
Neonatal Outcomes*:				
Caesarean Section (CS) or Instrumental Vaginal Birth (IVB) for fetal distress† (primary indication)				
CS for fetal distress (primary indication)				
IVB for fetal distress (primary indication)				
CS for other indications (primary indication not fetal distress)				
IVB for other indications (primary indication not fetal distress)				
CS or IVB – fetal distress was a factor				
CS – fetal distress was a factor				
IVB – fetal distress was a factor				
Spontaneous vaginal birth				
Maternal outcomes‡:				
Length of 1st and 2nd stage labour				
Post-partum hemorrhage (blood loss>1000ml)				
Post-partum hysterectomy				
Uterine rupture				
Maternal Intensive Care Unit admission				
Maternal mortality				

*GEE log-binomial

†CS for fetal distress defined as an emergency CS for fetal distress during labour as detected by fetal heart rate abnormalities based on the Royal Australian and New Zealand College of Obstetricians and Gynaecologists guidelines and/or abnormal fetal scalp pH or lactate levels. Instrumental vaginal birth for fetal distress defined as a vaginal birth using forceps or vacuum extraction for fetal distress.

‡Log-binomial model or t-test

Table 15: Subgroup analysis of the composite primary perinatal endpoint

	Sildenafil (N = ?)		**Placebo (N = ?)**		**Sildenafil vs Placebo: Relative Risk (95% CI)***	**P-value***	**Interaction P-value***
	N	*n (%)*	*N*	*n (%)*			
Small for gestational age (birthweight<10th centile for gestation):							x.xxx
Yes	xxx	xxx (xx.x)	xxx	xxx (xx.x)	x.xx (x.xx-x.xx)	x.xxx	
No	xxx	xxx (xx.x)	xxx	xxx (xx.x)	x.xx (x.xx-x.xx)	x.xxx	
Multiple pregnancy							x.xxx
Yes	xxx	xxx (xx.x)	xxx	xxx (xx.x)	x.xx (x.xx-x.xx)	x.xxx	
No	xxx	xxx (xx.x)	xxx	xxx (xx.x)	x.xx (x.xx-x.xx)	x.xxx	
Pre-eclampsia	xxx	xxx (xx.x)	xxx	xxx (xx.x)	x.xx (x.xx-x.xx)	x.xxx	
Yes	xxx	xxx (xx.x)	xxx	xxx (xx.x)	x.xx (x.xx-x.xx)	x.xxx	
No	xxx	xxx (xx.x)	xxx	xxx (xx.x)	x.xx (x.xx-x.xx)	x.xxx	
Preceding pregnancy CS	xxx	xxx (xx.x)	xxx	xxx (xx.x)	x.xx (x.xx-x.xx)	x.xxx	
Yes	xxx	xxx (xx.x)	xxx	xxx (xx.x)	x.xx (x.xx-x.xx)	x.xxx	
No	xxx	xxx (xx.x)	xxx	xxx (xx.x)	x.xx (x.xx-x.xx)	x.xxx	
Non-cephalic presentation	xxx	xxx (xx.x)	xxx	xxx (xx.x)	x.xx (x.xx-x.xx)	x.xxx	
Yes	xxx	xxx (xx.x)	xxx	xxx (xx.x)	x.xx (x.xx-x.xx)	x.xxx	
No	xxx	xxx (xx.x)	xxx	xxx (xx.x)	x.xx (x.xx-x.xx)	x.xxx	
Post-natal diagnosis of fetal anomaly	xxx	xxx (xx.x)	xxx	xxx (xx.x)	x.xx (x.xx-x.xx)	x.xxx	
Yes	xxx	xxx (xx.x)	xxx	xxx (xx.x)	x.xx (x.xx-x.xx)	x.xxx	
No	xxx	xxx (xx.x)	xxx	xxx (xx.x)	x.xx (x.xx-x.xx)	x.xxx	

*Log-binomial regression model